

(1) Mode choice — multinomial logit over the 7 modes:

$$P_{n,m} = \frac{\exp(U_{n,m})}{\sum_j \exp(U_{n,j})}$$

(2) Utility of mode m for agent n :

$$U_{n,m} = ASC_m + \boldsymbol{\beta}^\top \mathbf{x}_n + \beta_{GC} GC_{n,m}$$

(3) Generalized cost (minutes):

$$GC_{n,m} = t_{n,m} + \frac{60 c_{n,m}}{VOT_m(I_n)}$$

(4) Continuous income-dependent VOT (per agent):

$$VOT_m(I_n) = VOT_m^{ref} \cdot \text{clip}\left[\left(\frac{I_n}{I_{ref}}\right)^\lambda, 0.4, 2.5\right]$$

where $\mathbf{x}_n$: socio-demographics (β estimated from the RTS);
 $c_{n,m}$: cost \$ = distance × \$0.20 + toll (full on car-driver, half shared-ride); I_n : monthly income; I_{ref} = \$7,018/mo; $\lambda = 0.6$;
 VOT^{ref} = car 30 / shared-ride 40 / transit 15 \$/h.

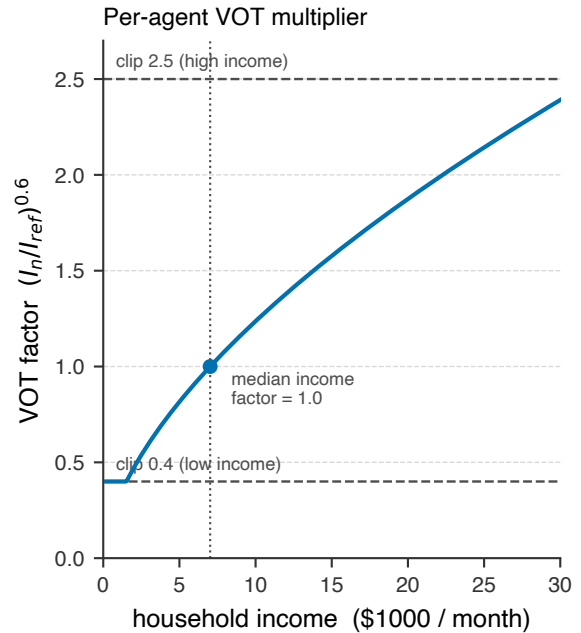


Figure. Income-dependent value-of-time (VOT) mode choice. The 7-mode generalized-cost logit (Chayan & Cirillo, 2024) is extended with a continuous per-agent income term: VOT scales with $(I/I_{ref})^{0.6}$, clipped to $[0.4, 2.5]$ and equal to 1.0 at the region-median income (\$7,018/mo). Lower income $\rightarrow$ lower VOT $\rightarrow$ the toll's dollar cost weighs more in utility $\rightarrow$ a larger mode shift away from the tolled car; at the median the base split is preserved.